

4. Find the acceleration of the nuclei at the present time step i , using Newton's law of gravity,

$$A_{1,x}(i) = \frac{GM_2}{s^3(i)} [X_2(i) - X_1(i)] \quad (\text{M.7})$$

and

$$A_{2,x}(i) = \frac{GM_1}{s^3(i)} [X_1(i) - X_2(i)], \quad (\text{M.8})$$

where $s(i)$ is the separation of the nuclei at time step i ,

$$s(i) = \sqrt{[X_1(i) - X_2(i)]^2 + [Y_1(i) - Y_2(i)]^2 + [Z_1(i) - Z_2(i)]^2 + s_f^2}. \quad (\text{M.9})$$

5. Find the velocity of the nuclei at $i + 1/2$,

$$V_{1,x}\left(i + \frac{1}{2}\right) = V_{1,x}\left(i - \frac{1}{2}\right) + A_{1,x}(i)\Delta t, \quad (\text{M.10})$$

and similarly for $V_{2,x}(i + 1/2)$.

6. Find the position of the nuclei at the next time step $i + 1$, using

$$X_1(i + 1) = X_1(i) + V_{1,x}\left(i + \frac{1}{2}\right)\Delta t, \quad (\text{M.11})$$

and similarly for $X_2(i + 1)$.

The procedure is the same for the y and z components. By repeatedly applying this prescription, it is possible to follow the motions of the nuclei and star(s).

The target galaxy (M_1) is initially placed at rest at the origin. You will be asked to provide the initial position and velocity of the intruder galaxy (M_2), its mass (as a fraction of M_1), and the number of stars around the target galaxy. After each time step, the results are displayed as two graphs showing the positions of the nuclei and stars on the x - y and x - z planes.

You will note in the source code that the program uses a special system of units to speed the calculations. The masses are in units of $2 \times 10^{10} M_\odot$. When the program assigns the target galaxy a mass of 5 in these units, its mass is $10^{11} M_\odot$. The unit of time is 1.2 million years. This is also the value used for the time interval, Δt , so (in these units) $\Delta t = 1$. As a result, Δt does not appear explicitly in the program. (It would just multiply the term involved by 1 and waste computer time.) The unit of distance is 500 pc, and so the unit of velocity is $(500 \text{ pc})/(1.2 \text{ million years}) \simeq 400 \text{ km s}^{-1}$. By design, in these units the gravitational constant $G = 1$; thus G does not appear explicitly in the program either.

The source code for Galaxy, along with executable versions, is available for download from the resources tab on the following webpage: www.cambridge.org/astrophysics.

APPENDIX

N

WMAP Data

“Best” Cosmological Parameters ^a				
Description	Text Symbol	Value	+ uncertainty	− uncertainty
Total density	Ω_0	1.02	0.02	0.02
Equation of state of quintessence ^b	w	< -0.78	95% CL	
Dark energy density	$\Omega_{\Lambda,0}$	0.73	0.04	0.04
Baryon density	$\Omega_{b,0}h^2$	0.0224	0.0009	0.0009
Baryon density	$\Omega_{b,0}$	0.044	0.004	0.004
Baryon density (m^{-3})	$n_{b,0}$	0.25	0.01	0.01
Matter density	$\Omega_{m,0}h^2$	0.135	0.008	0.009
Matter density	$\Omega_{m,0}$	0.27	0.04	0.04
Light neutrino density (m^{-3})	$\Omega_{\nu,0}h^2$	< 7600	95% CL	
CMB temperature (K) ^c	T_0	2.725	0.002	0.002
CMB photon density (m^{-3}) ^d	$n_{\gamma,0}$	4.104×10^8	0.009×10^8	0.009×10^8
Baryon-to-photon ratio	η_0	6.1×10^{-10}	0.3×10^{-10}	0.2×10^{-10}
Baryon-to-matter ratio	$\Omega_{b,0}\Omega_{m,0}^{-1}$	0.17	0.01	0.01
Redshift at decoupling	z_{dec}	1089	1	1
Thickness of decoupling (FWHM)	Δz_{dec}	195	2	2
Hubble constant	h	0.71	0.04	0.03
Age of universe (Gyr)	t_0	13.7	0.2	0.2
Age at decoupling (kyr)	t_{dec}	379	8	7
Age at reionization (Myr, 95% CL)	t_r	180	220	80
Decoupling time interval (kyr)	Δt_{dec}	118	3	2
Redshift of matter–energy equality	$z_{r,m}$	3233	194	210
Reionization optical depth	τ	0.17	0.04	0.04
Redshift at reionization (95% CL)	z_r	20	10	9
Sound horizon at decoupling (deg)	θ_A	0.598	0.002	0.002
Angular size distance (Gpc)	d_A	14.0	0.2	0.3
Acoustic scale ^e	ℓ_A	301	1	1
Sound horizon at decoupling (Mpc) ^f	r_s	147	2	2

^a All data from Bennett et al., *Ap. J. S.*, 148, 1, 2003.

^b CL means “confidence level.”

^c From COBE (Mather et al., *Ap. J.*, 512, 511, 1999).

^d Derived from COBE (Mather et al., *Ap. J.*, 512, 511, 1999).

^e $\ell_A = \pi\theta_A^{-1}$ for θ_A in radians.

^f $\theta_A = r_s d_A^{-1}$ for θ_A in radians.

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